

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - 21

Roll No. : S037	Name:Himanshu
Class : SYIT	Batch : 01
Date of Assignment : 18-8-26	Date/Time of Submission :18-8-26

Q.Join two CSV files in Spark DataFrames based on a common column and write the output to a file.

Code:

Scala Version 2.12.18

```
import org.apache.spark.sql.SparkSession

object Demo extends App {
  val s = SparkSession.builder()
    .appName("JoinTwoCSV")
    .master("local[*]")
    .getOrCreate()

  val d1 = s.read
    .option("header", "true")
    .csv("file1.csv")

  val d2 = s.read
    .option("header", "true")
    .csv("file2.csv")

  val j = d1.join(d2, Seq("id"), "inner")

  j.show()
```

```
j.write.option("header", "true")

s.stop()
}
```

Output;

```
+---+-----+-----+
| id|  name|score|
+---+-----+-----+
|  1|  Rani|   85|
|  2|Sachin|   40|
|  3| Priya|   78|
+---+-----+-----+
```

Practical 22

22) Set up Apache Spark locally and count the frequency of words in a text file.

Code:

```
import org.apache.spark.sql.SparkSession

object Demo extends App
{
    val s = SparkSession.builder()
        .appName("WordCountExample")
        .master("local[*]")
        .getOrCreate()

    val r = s.sparkContext.textFile("input.txt")
    val w = r.flatMap(line => line.split("\\s+"))
    val wo = w.map(word => (word, 1))
    val wc = wo.reduceByKey(_ + _)

    wc.collect().foreach(println)
    s.stop()
}
```

```
}
```

Output:

```
(is,2)
(easy,1)
(powerful,1)
(Spark,2)
```

Practical 23

23) Filter rows in a CSV file using Spark DataFrames where a numeric column exceeds a threshold.

Code:

```
import org.apache.spark.sql.SparkSession

object Demo extends App {
  val spark = SparkSession.builder()
    .appName("FilterCSVExample")
    .master("local[*]")
    .getOrCreate()

  import spark.implicits._

  val df = spark.read
    .option("header", "true")
    .option("inferSchema", "true")
    .csv("src/main/resources/superstore.csv")

  val f = df.filter($"Sales" > 5000)

  println("Filtered Data:")
  f.show()
  spark.stop()
}
```

Output:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	TEC-MA-10000822
510	CA-2015-145352	3/16/2015	3/22/2015	Standard Class	CM-12385	Christopher Martinez	Consumer	United States	Atlanta	Georgia	30318	South	OFF-BI-10003527
684	US-2017-168116	11/4/2017	11/4/2017	Same Day	GT-14635	Grant Thornton	Corporate	United States	Burlington	North Carolina	27217	South	TEC-MA-10004125
2506	CA-2014-143917	7/25/2014	7/27/2014	Second Class	KL-16645	Ken Lonsdale	Consumer	United States	San Francisco	California	94122	West	OFF-SU-10000151
2624	CA-2017-127180	10/22/2017	10/24/2017	First Class	TA-21385	Tom Ashbrook	Home Office	United States	New York City	New York	10024	East	TEC-CO-10004722
2698	CA-2014-145317	3/18/2014	3/23/2014	Standard Class	SM-20320	Sean Miller	Home Office	United States	Jacksonville	Florida	32216	South	TEC-MA-10002412
4099	CA-2014-116904	9/23/2014	9/28/2014	Standard Class	SC-20095	Sanjit Chand	Consumer	United States	Minneapolis	Minnesota	55407	Central	OFF-BI-10001120
4191	CA-2017-166709	11/17/2017	11/22/2017	Standard Class	HL-15040	Hunter Lopez	Consumer	United States	Newark	Delaware	19711	East	TEC-CO-10004722
4278	US-2016-107440	4/16/2016	4/20/2016	Standard Class	BS-11365	Bill Shonely	Corporate	United States	Lakewood	New Jersey	8701	East	TEC-MA-10001047
6341	CA-2017-143112	10/5/2017	10/9/2017	Standard Class	TS-21370	Todd Sumrall	Corporate	United States	New York City	New York	10035	East	TEC-MA-10001047
6426	CA-2016-143714	5/23/2016	5/27/2016	Standard Class	CC-12370	Christopher Conant	Consumer	United States	Philadelphia	Pennsylvania	19120	East	TEC-CO-10004722
6521	CA-2017-138289	1/16/2017	1/18/2017	Second Class	AR-10540	Andy Reiter	Consumer	United States	Jackson	Michigan	49201	Central	OFF-BI-10004995
6627	CA-2014-145541	12/14/2014	12/21/2014	Standard Class	TB-21400	Tom Boeckenhauer	Consumer	United States	New York City	New York	10024	East	TEC-MA-10001127
6827	CA-2016-118689	10/2/2016	10/9/2016	Standard Class	TC-20080	Tamara Chand	Corporate	United States	Lafayette	Indiana	47905	Central	TEC-CO-10004722
7667	US-2016-140158	10/4/2016	10/8/2016	Standard Class	DR-12940	Daniel Raglin	Home Office	United States	Providence	Rhode Island	2908	East	TEC-CO-10001449
8154	CA-2017-140151	3/23/2017	3/25/2017	First Class	RB-19360	Raymond Buch	Consumer	United States	Seattle	Washington	98115	West	TEC-CO-10004722
8489	CA-2016-158841	2/2/2016	2/4/2016	Second Class	SE-20110	Sanjit Engle	Consumer	United States	Arlington	Virginia	22204	South	TEC-MA-10001127
8859	CA-2017-135909	10/13/2017	10/20/2017	Standard Class	JW-15220	Jane Wacoi	Corporate	United States	Sacramento	California	95823	West	OFF-BI-10003527
9040	CA-2016-117121	12/17/2016	12/21/2016	Standard Class	AB-10105	Adrian Barton	Consumer	United States	Detroit	Michigan	48205	Central	OFF-BI-10000545

Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
Technology	Machines	Lexmark MX611dhe ...	8159.952	8	0.4	-1359.992
Office Supplies	Binders	Fellowes PB500 EL...	6354.95	5	0	3177.475
Technology	Machines	Cubify CubeX 3D P...	7999.98	4	0.5	-3839.9904
Office Supplies	Supplies	High Speed Automa...	8187.65	5	0	327.506
Technology	Copiers	Canon imageCLASS ...	11199.968	4	0.2	3919.9888
Technology	Machines	Cisco TelePresenc...	22638.48	6	0.5	-1811.0784
Office Supplies	Binders	Ibico EPK-21 Elec...	9449.95	5	0	4630.4755
Technology	Copiers	Canon imageCLASS ...	10499.97	3	0	5039.9856
Technology	Machines	3D Systems Cube P...	9099.93	7	0	2365.9818
Technology	Machines	3D Systems Cube P...	5199.96	4	0	1351.9896
Technology	Copiers	Canon imageCLASS ...	8399.976	4	0.4	1119.9968
Office Supplies	Binders	GBC DocuBind P400...	5443.96	4	0	2504.2216
Technology	Machines	"HP Designjet T52...	6999.96	4	0	2239.9872
Technology	Copiers	Canon imageCLASS ...	17499.95	5	0	8399.976
Technology	Copiers	Hewlett Packard L...	5399.91	9	0	2591.9568
Technology	Copiers	Canon imageCLASS ...	13999.96	4	0	6719.9808
Technology	Machines	"HP Designjet T52...	8749.95	5	0	2799.984
Office Supplies	Binders	Fellowes PB500 EL...	5083.96	5	0.2	1906.485
Office Supplies	Binders	GBC Ibimaster 500...	9892.74	13	0	4946.37

Practical 24.

24) Perform a group-by operation in Spark DataFrames to compute the average of each group.

Code:

```
import org.apache.spark.sql.SparkSession
```

```
import org.apache.spark.sql.functions._
```

```
object Demo extends App {
```

```
    val spark = SparkSession.builder()
```

```

    .appName("AverageSales")
    .master("local[*]")
    .getOrCreate()

val df = spark.read
    .option("header", "true")
    .option("inferSchema", "true")
    .csv("src/main/resources/superstore.csv")

val result = df.groupBy("Category")
    .agg(avg("Sales").alias("Average Sales"))

println("Average Sales by Category:")
result.show()

spark.stop()
}

```

Output:

```

20/06/20 11:00:40 INFO Output:
Average Sales by Category:

```

```

+-----+-----+
|      Category|  Average Sales|
+-----+-----+
|Office Supplies|121.69225531914947|
|      Furniture| 353.4459311957568|
|    Technology| 454.5405475802047|
+-----+-----+

```